SVM

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Imports

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.cm as cm
from sklearn.model_selection import train_test_split , cross_val_score, Stratifi
from sklearn.svm import SVC
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, ConfusionMatrixDisplay, confusion_ma
from sklearn.model_selection import cross_val_score
import seaborn as sns
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_sc
from sklearn.metrics import roc_curve, auc
```

Helpfull Code

Load Data

In [2]:
```
spaceship_data = pd.read_csv('final_data.csv')
```

split to x and y

In [3]:
```
y = spaceship_data["Transported"].astype(int)
x = spaceship_data.drop(columns=["Transported"])
```

Train-Test Split

In [4]:
```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

In [5]:
```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

בדיקת ערכי ההייפר פרמטרים

In [6]:
```
results = []
best_score = 0
best_params_svm = None
```

In [7]:
```
for C in [0.1, 1, 10, 100, 1000]:
    for degree in [1, 2, 3, 4, 5]:

        svm_pipe = Pipeline([
            ("scaler", StandardScaler()),
            ("svm", SVC(
                kernel="poly",
                C=C,
                degree=degree,
                random_state=42
            ))
        ])

        fold_scores = cross_val_score(
            svm_pipe,
            x_train,
            y_train,
            cv=cv,
            scoring="accuracy",
            n_jobs=-1
        )

        mean_score = fold_scores.mean()

        print(f"\nSVM POLY | C={C}, degree={degree}")
        print("Fold accuracies (%):", np.round(100 * fold_scores, 2))
        print(f"Mean CV accuracy: {100 * mean_score:.2f}%")

        results.append({
            "C": C,
            "degree": degree,
            "cv_mean": mean_score,
            "cv_std": fold_scores.std()
        })

        if mean_score > best_score:
            best_score = mean_score
            best_params_svm = {
                "C": C,
                "degree": degree
            }
```

```
SVM POLY | C=0.1, degree=1
Fold accuracies (%): [75.4  75.99 77.87 75.24 76.84]
Mean CV accuracy: 76.27%

SVM POLY | C=0.1, degree=2
Fold accuracies (%): [73.21 73.14 74.36 73.05 74.07]
Mean CV accuracy: 73.57%

SVM POLY | C=0.1, degree=3
Fold accuracies (%): [75.04 75.91 76.84 75.53 75.68]
Mean CV accuracy: 75.80%

SVM POLY | C=0.1, degree=4
Fold accuracies (%): [71.75 72.77 73.7  73.41 72.9 ]
Mean CV accuracy: 72.91%

SVM POLY | C=0.1, degree=5
Fold accuracies (%): [69.78 70.88 72.61 71.88 71.95]
Mean CV accuracy: 71.42%

SVM POLY | C=1, degree=1
Fold accuracies (%): [76.57 77.45 80.42 76.77 77.65]
Mean CV accuracy: 77.77%

SVM POLY | C=1, degree=2
Fold accuracies (%): [76.2  76.28 79.55 76.63 76.63]
Mean CV accuracy: 77.06%

SVM POLY | C=1, degree=3
Fold accuracies (%): [77.01 77.01 78.09 77.28 78.09]
Mean CV accuracy: 77.49%

SVM POLY | C=1, degree=4
Fold accuracies (%): [76.42 75.77 77.72 76.7  77.21]
Mean CV accuracy: 76.76%

SVM POLY | C=1, degree=5
Fold accuracies (%): [75.11 75.55 76.11 75.31 76.48]
Mean CV accuracy: 75.71%

SVM POLY | C=10, degree=1
Fold accuracies (%): [78.61 78.32 81.52 77.94 78.38]
Mean CV accuracy: 78.95%

SVM POLY | C=10, degree=2
Fold accuracies (%): [78.18 77.81 79.99 77.21 77.79]
Mean CV accuracy: 78.19%

SVM POLY | C=10, degree=3
Fold accuracies (%): [77.81 79.2  80.64 78.23 79.33]
Mean CV accuracy: 79.04%

SVM POLY | C=10, degree=4
Fold accuracies (%): [77.66 76.28 78.67 76.77 77.14]
Mean CV accuracy: 77.30%

SVM POLY | C=10, degree=5
Fold accuracies (%): [77.3  76.86 77.57 77.36 77.5 ]
Mean CV accuracy: 77.32%
```

```
SVM POLY | C=100, degree=1
Fold accuracies (%): [78.69 78.39 81.3  78.23 78.38]
Mean CV accuracy: 79.00%

SVM POLY | C=100, degree=2
Fold accuracies (%): [78.83 78.91 80.72 78.52 78.52]
Mean CV accuracy: 79.10%

SVM POLY | C=100, degree=3
Fold accuracies (%): [78.1  78.25 79.18 77.87 78.09]
Mean CV accuracy: 78.30%

SVM POLY | C=100, degree=4
Fold accuracies (%): [77.01 76.93 78.31 76.7  76.19]
Mean CV accuracy: 77.03%

SVM POLY | C=100, degree=5
Fold accuracies (%): [77.59 77.01 77.14 77.94 77.28]
Mean CV accuracy: 77.39%

SVM POLY | C=1000, degree=1
Fold accuracies (%): [78.69 78.39 81.3  78.23 78.45]
Mean CV accuracy: 79.01%

SVM POLY | C=1000, degree=2
Fold accuracies (%): [78.98 78.91 80.28 78.96 79.04]
Mean CV accuracy: 79.23%

SVM POLY | C=1000, degree=3
Fold accuracies (%): [76.28 77.96 78.23 76.92 76.48]
Mean CV accuracy: 77.17%

SVM POLY | C=1000, degree=4
Fold accuracies (%): [73.94 75.18 75.97 75.38 73.85]
Mean CV accuracy: 74.86%

SVM POLY | C=1000, degree=5
Fold accuracies (%): [77.15 76.06 76.04 76.48 75.97]
Mean CV accuracy: 76.34%
```

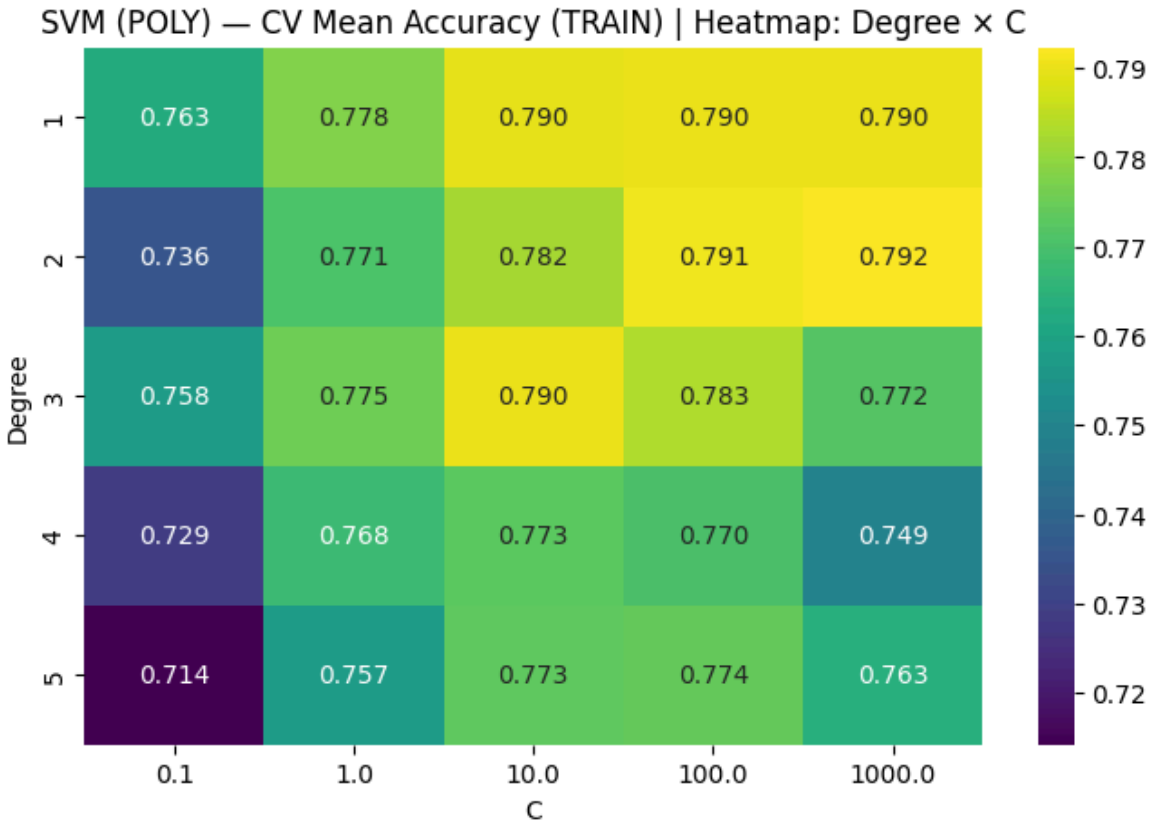
הצגת הטובים ביותר ותוצאותיהם

In [8]:
```
print("\nBest SVM params:", best_params_svm)
print(f"Best CV accuracy: {100 * best_score:.2f}%")

results_df = pd.DataFrame(results).sort_values("cv_mean", ascending=False)

display(results_df.head(10).round(4))
```

```
Best SVM params: {'C': 1000, 'degree': 2}
Best CV accuracy: 79.23%
```

	C	degree	cv_mean	cv_std
21	1000.0	2	0.7923	0.0052
16	100.0	2	0.7910	0.0082
12	10.0	3	0.7904	0.0098
20	1000.0	1	0.7901	0.0115
15	100.0	1	0.7900	0.0116
10	10.0	1	0.7895	0.0130
17	100.0	3	0.7830	0.0046
11	10.0	2	0.7819	0.0095
5	1.0	1	0.7777	0.0139
7	1.0	3	0.7749	0.0049

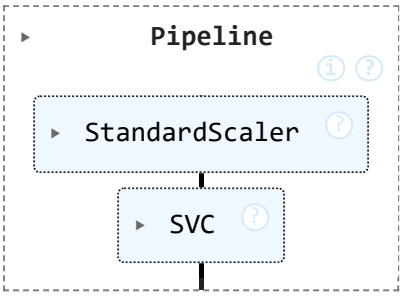
In [9]:
```
heatmap_data = results_df.pivot(index="degree", columns="C", values="cv_mean")

plt.figure(figsize=(8,5))
sns.heatmap(heatmap_data, annot=True, fmt=".3f", cmap="viridis")
plt.title("SVM (POLY) — CV Mean Accuracy (TRAIN) | Heatmap: Degree × C")
plt.xlabel("C")
plt.ylabel("Degree")
plt.show()
```



Best SVM

In [10]:
```
best_SVM = Pipeline([
    ("scaler", StandardScaler()),
    ("svm", SVC(
        kernel="poly",
        C=best_params_svm["C"],
        degree=best_params_svm["degree"],
        random_state=42
    ))
])

best_SVM.fit(x_train, y_train)
```

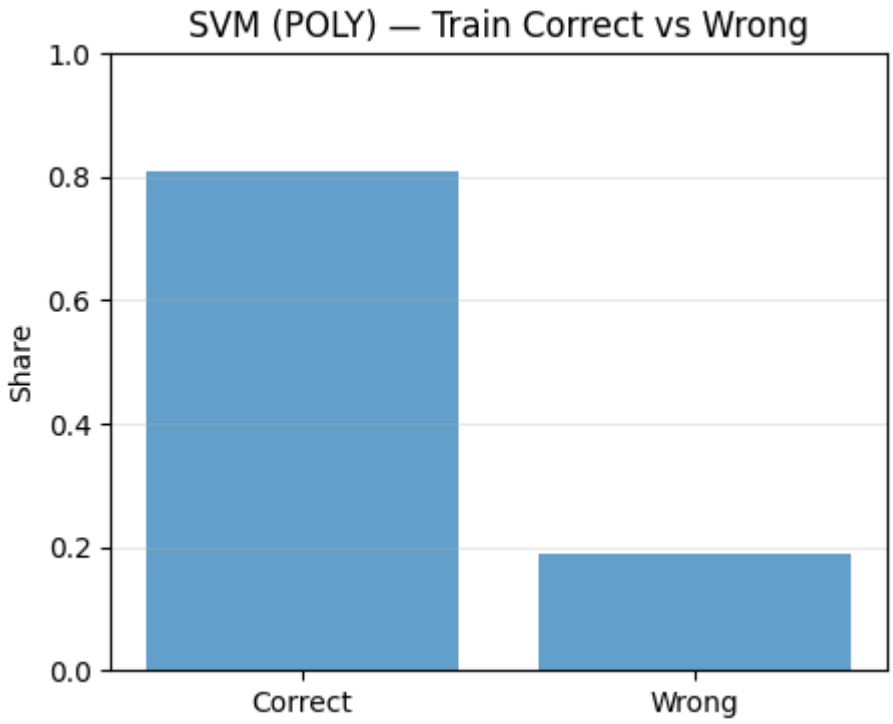
Out[10]:



Results

In [11]:
```
y_pred_train = best_SVM.predict(x_train)
```

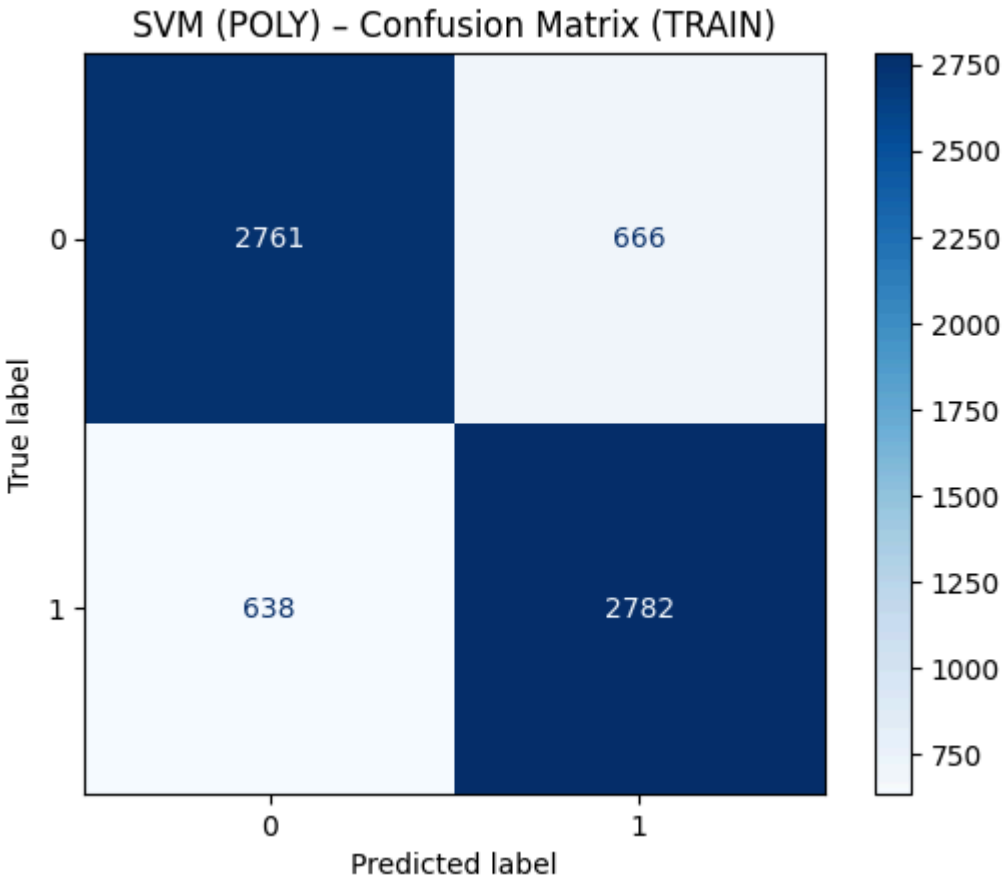
In [12]:
```
train_acc = accuracy_score(y_train, y_pred_train)
train_err = 1 - train_acc

plt.figure(figsize=(5,4))
plt.bar(["Correct", "Wrong"], [train_acc, train_err], alpha=0.7)
plt.ylim(0, 1)
plt.ylabel("Share")
plt.title("SVM (POLY) — Train Correct vs Wrong")
plt.grid(axis="y", alpha=0.3)
plt.show()


print("="*40)
print("BEST SVM (POLY) — TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc*100:.2f}%")
print("="*40)
```




```
========================================
BEST SVM (POLY) — TRAIN SET PERFORMANCE
========================================
Accuracy : 80.96%
========================================
```

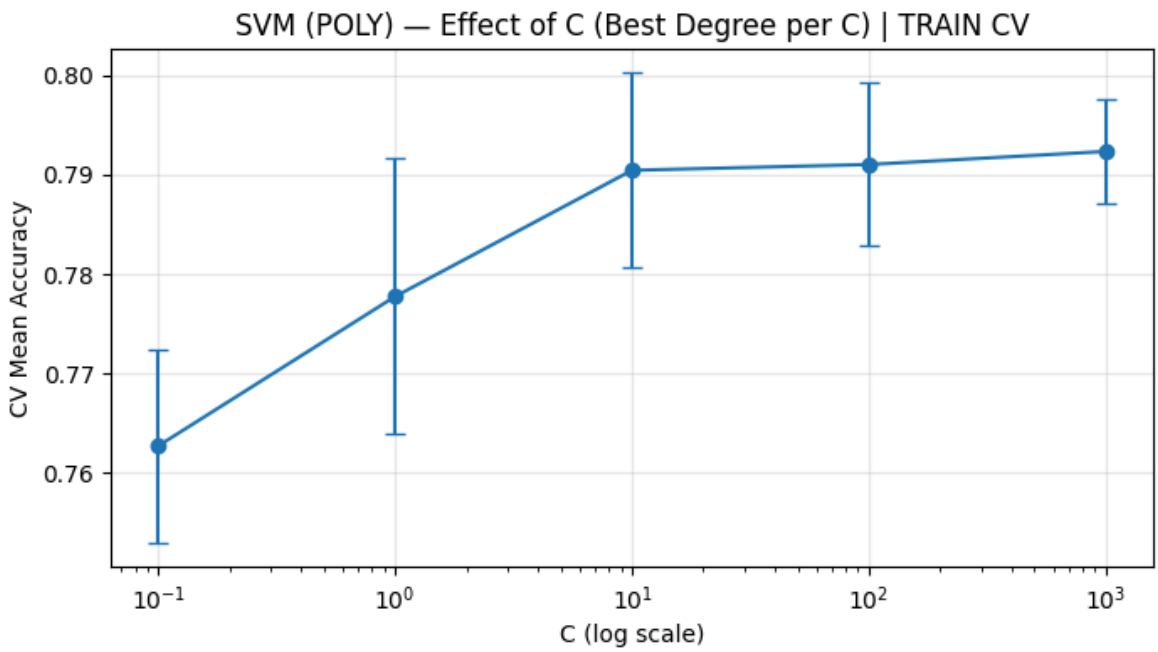
confusion_matrix

In [13]:
```
cm_train = confusion_matrix(y_train, y_pred_train)

# display
disp = ConfusionMatrixDisplay(confusion_matrix=cm_train)
disp.plot(cmap="Blues")

plt.title("SVM (POLY) - Confusion Matrix (TRAIN)")
plt.show()

print(f"Train Accuracy: {100 * accuracy_score(y_train, y_pred_train):.2f}%")
```

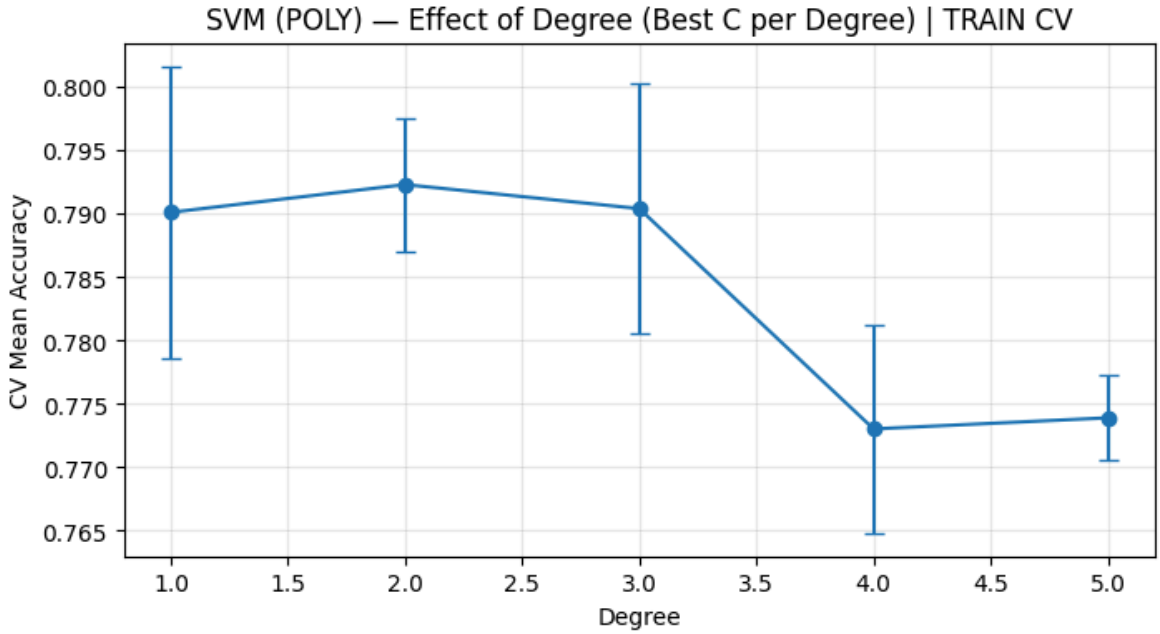



```
Train Accuracy: 80.96%
```

In [14]:
```
plot_c = (
    results_df
    .sort_values(["C", "cv_mean"], ascending=[True, False])
    .groupby("C", as_index=False)
    .head(1)
    .sort_values("C")
)

plt.figure(figsize=(8,4))
plt.errorbar(plot_c["C"], plot_c["cv_mean"], yerr=plot_c["cv_std"], fmt="o-", ca
plt.xscale("log")
plt.title("SVM (POLY) — Effect of C (Best Degree per C) | TRAIN CV")
plt.xlabel("C (log scale)")
plt.ylabel("CV Mean Accuracy")
plt.grid(alpha=0.3)
plt.show()
```



In [15]:
```
plot_deg = (
    results_df
    .sort_values(["degree", "cv_mean"], ascending=[True, False])
    .groupby("degree", as_index=False)
    .head(1)
    .sort_values("degree")
)

plt.figure(figsize=(8,4))
plt.errorbar(plot_deg["degree"], plot_deg["cv_mean"], yerr=plot_deg["cv_std"], f
plt.title("SVM (POLY) — Effect of Degree (Best C per Degree) | TRAIN CV")
plt.xlabel("Degree")
plt.ylabel("CV Mean Accuracy")
plt.grid(alpha=0.3)
plt.show()
```



In [16]:
```
acc = accuracy_score(y_train, y_pred_train)
prec = precision_score(y_train, y_pred_train)
```

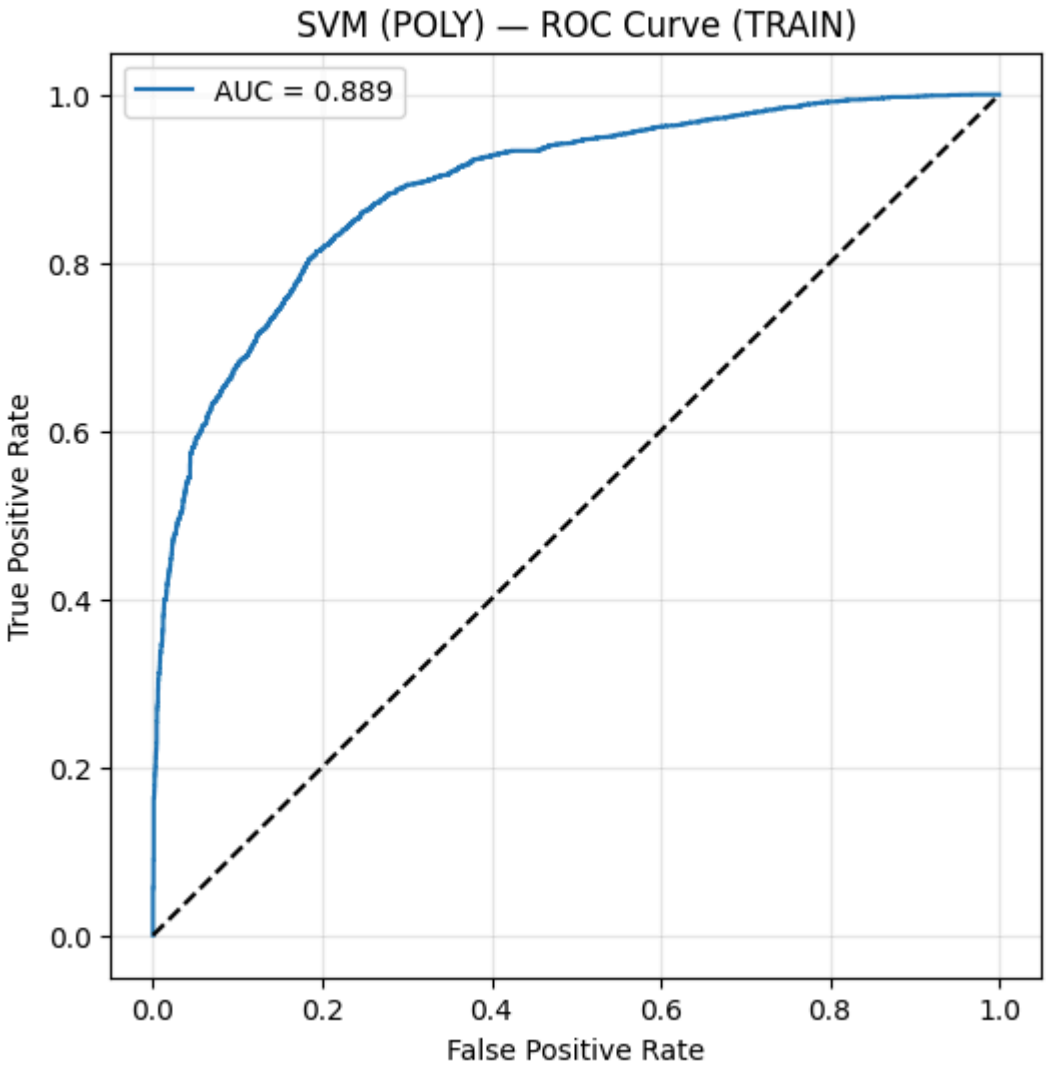
```
rec = recall_score(y_train, y_pred_train)
f1 = f1_score(y_train, y_pred_train)

metrics_df = pd.DataFrame([{
    "model": "SVM_POLY",
    "accuracy": acc,
    "precision": prec,
    "recall": rec,
    "f1_score": f1
}])

display(metrics_df.round(4))

print("\nClassification Report (TRAIN):")
print(classification_report(y_train, y_pred_train))
```

	model	accuracy	precision	recall	f1_score
0	SVM_POLY	0.8096	0.8068	0.8135	0.8101

```
Classification Report (TRAIN):
              precision    recall  f1-score   support

           0       0.81      0.81      0.81      3427
           1       0.81      0.81      0.81      3420

    accuracy                           0.81      6847
   macro avg       0.81      0.81      0.81      6847
weighted avg       0.81      0.81      0.81      6847
```

In [17]:

```
# SVM score for ROC (works without predict_proba)
y_score_train = best_SVM.decision_function(x_train)

fpr, tpr, thresholds = roc_curve(y_train, y_score_train)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6,6))
plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}")
plt.plot([0,1], [0,1], "k--")

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("SVM (POLY) — ROC Curve (TRAIN)")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```


SVM (POLY) — ROC Curve (TRAIN)
AUC = 0.889
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0